

Review article

Satellite cell activation induced by aerobic muscle adaptation in response to endurance exercise in humans and rodents



Phablo Abreu^{a,b,*}, Sávio Victor Diógenes Mendes^b, Vânia Marilande Ceccatto^b, Sandro Massao Hirabara^{a,c}

^a Institute of Biomedical Sciences, University of São Paulo, SP, Brazil

^b Institute of Biomedical Sciences, State University of Ceará, CE, Brazil

^c Institute of Physical Activity Sciences and Sports, Cruzeiro do Sul University, São Paulo, SP, Brazil

ARTICLE INFO

Article history:

Received 29 July 2016

Received in revised form 31 October 2016

Accepted 21 November 2016

Available online 22 November 2016

Keywords:

Satellite cells activation

Skeletal muscle

Endurance exercise

ABSTRACT

Although the requirement of satellite cells activation and expansion following injury, mechanical load or growth stimulus provoked by resistance exercise has been well established, their function in response to aerobic exercise adaptation remains unclear. A clear relationship between satellite cell expansion in fiber-type specific myosin heavy chain and aerobic performance has been related, independent of myonuclear accretion or muscle growth. However, the trigger for this activation process is not fully understood yet and it seems to be a multi-faceted and well-orchestrated process. Emerging *in vitro* studies suggest a role for metabolic pathways and oxygen availability for satellite cell activation, modulating the self-renewal potential and cell fate control. The goal of this review is to describe and discuss the current knowledge about the satellite cell activation and expansion in response to aerobic exercise adaptation in human and rodent models. Additionally, findings about the *in vitro* metabolic control, which seems to be involved in the satellite cell activation and cell fate control, are presented and discussed.

© 2016 Elsevier Inc. All rights reserved.

Contents

1. Introduction	33
2. Physiology of the muscle satellite cells	34
3. Metabolic regulation <i>in vitro</i> of satellite cell activation	34
4. Satellite cell activation in response to aerobic exercise adaptation: evidence in rodent models	35
5. Satellite cell activation in response to aerobic exercise adaptation: evidence in human.	36
6. Conclusion	37
Conflicts of interest statement	38
Acknowledgments	38
References	38

1. Introduction

Skeletal muscle is composed by regular multinucleated myofibers that possess the contractile machinery to generate force and activity. After the discovery of the satellite cells by Alexander Mauro in 1961 [1], their activation following diseases, toxins, injury, mechanical load, growth stimulus, and resistance exercise has been extensively studied. However, the function of satellite cells induced by aerobic exercise

adaptation remains unclear and largely unexplored [2–14]. Satellite cells expansion occurs in a fiber-type specific manner, independent of myonuclear accretion and myofiber fusion, suggesting a correlation between satellite cell activation and aerobic muscle performance [6,15–19].

Interestingly, previous *in vitro* studies appoint out a role for metabolic pathways and oxygen availability in satellite cell activation, which can modulate self-renewal potential and cell fate control. Quiescent satellite cells have low metabolic activity; however, after activation, these cells have several metabolic alterations associated with the expression of specific bioenergetics proteins [20].

The goal of this review is to describe and discuss the current knowledge about the satellite cell activation in response to aerobic exercise

* Corresponding author: Department of Physiology and Biophysics, Institute de Biomedical Sciences, University of São Paulo, Av. Prof. Lineu Prestes 1524, Butantã, São Paulo, SP 05508-000, Brazil.

E-mail address: phablo@icb.usp.br (P. Abreu).

adaptation in human and rodent models. Additionally, findings about the *in vitro* metabolic activity, which seems to be involved in the satellite cell activation and cell fate control, are presented and discussed. Comprehension of the molecular basis that control the function of satellite cells constitutes an interesting research area, aiming to the development of potential muscle stem cell therapeutic interventions for muscle regeneration in a wide variety of neuromuscular illnesses, which are characterized by intense skeletal muscle wasting, including dystrophy, cachexia, and age-related sarcopenia.

2. Physiology of the muscle satellite cells

Satellite cells in adult skeletal muscle are small mononucleated muscle stem cells located under the basal lamina of myofibers, representing 2–10% of the total muscle myonuclei (2×10^5 to 1×10^6 cells/g muscle) [21]. In normal (non-injured) muscle, satellite cells are quiescent (G0 phase) and transcriptionally inactive [21–23]. The balance in quiescence and activation states is crucial to properly ensure the life-long maintenance of skeletal muscle tissue and to guarantee the constant permanence of these cells even after severe muscle injuries [24].

Satellite cells are characterized by expressing the paired box protein Pax7, which is expressed in multiple species, including mouse [25], human and monkey [26], zebrafish [27], chick [28], frog [29], pig [30], and salamander [31]. Stem cells in adult skeletal muscle can be identified by fluorescence microscopy through specific biomarkers, such as Pax3 and 7 expression [32,33], myogenic regulatory factor Myf5 [34], cell adhesion protein M-cadherin [35], cluster of differentiation protein CD34 [36], tyrosine receptor kinase c-Met [37], cell surface attachment receptor α 7-intergrin [38], transmembrane heparan sulfate proteoglycans syndecan-3 and syndecan-4 [39], chemokine receptor CXCR4 [40], caveolae-forming protein caveolin-1, nuclear envelope proteins lamin A/C and emerin [38], calcitonin receptor [41], and homeobox transcription factor Barx2 [42].

Previous studies using electronic microscopy showed that the number of satellite cells in MyHC type I fibers, e.g. *soleus* and *diaphragm*, is strictly associated with the numbers of capillaries [43]. Similarly, Gibson and Schultz [44] demonstrated that the satellite cell content is higher in oxidative muscle, e.g. *soleus*, than glycolytic muscle, e.g. *extensor digitorum longus* (EDL), from adult rats.

Since their discovery, the research field of muscle stem cells has immensely grown in the last years, mainly due to the development of sophisticated techniques of molecular and cell biology that have significantly contributed for the elucidation of the molecular and cellular mechanisms (identification of intrinsic and extrinsic factors) in different states of satellite cells and skeletal muscle plasticity.

3. Metabolic regulation *in vitro* of satellite cell activation

Previous studies have been proposed metabolism signals control muscle satellite cell function and cell fate *in vitro*. For example, the signaling pathways regulated by the integration of local and systemic factors that influence the cell metabolism comprehend that activated by insulin and insulin-like growth factor 1 (IGF-1), including the serine/threonine kinase Akt (also known as protein kinase B or PKB)/phosphoinositide 3-kinase (PI3K), forkhead transcription factor (FOXO), mammalian target of rapamycin (mTOR), AMP-activated protein kinase (AMPK), peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α), hypoxia-inducible transcription factor (Hif1 α), nicotinamide adenine dinucleotide (NAD), and sirtuin (SIRT) pathways.

The content of substrate, oxygen availability, as well as the cell energy demand, is important for the determination of the metabolic pathway for adenosine triphosphate (ATP) generation. In conditions of high availability of oxygen, ATP is mainly generated *via* oxidative phosphorylation by mitochondria. In the other hand, under conditions of low oxygen tension or high-energy demand, ATP is generated *via* glycolysis

in the cytoplasm. This last one provides glycolytic intermediates for the biosynthesis of new essential macromolecules for expansion cells [45–47]. More recently it has been established that metabolic state of the satellite cells can directly or indirectly affect the activity of chromatin-modifying enzymes, which modulate satellite cell fate [48,49].

The metabolic regulation for satellite cell activation could occur in response to several conditions, including hypoxia, oxidative stress, inflammation, exercise, and aging, being a coordinated and intricate process for muscle-specific gene expression pattern at transcriptional and post-transcriptional levels [50].

Inactivated muscle satellite cells have small metabolic rate at quiescent state, but after entrance into the cell cycle, these cells suffer several metabolic changes, shifting from oxidative to glycolytic metabolism. This process is accompanied by a reduction of the NAD⁺/NADH ratio and, consequently, activity of SIRT1-mediated deacetylation of histone H4 (H4K16ac), resulting in activation of myogenic program [19] (Fig. 1).

Several metabolic sensors and signaling pathways have been well recognized to modulate cell function in a well-controlled process. Low oxygen tension (hypoxia) seems to be a crucial factor for activation of stem cells, playing a crucial role in the proliferation and self-renewal of the muscle satellite cells. Interestingly, previous studies revealed that hypoxia maintains myoblasts in an undifferentiated state *in vitro* by inhibiting myogenic differentiation through activation of the Notch or PI3K signaling pathway [51,52]. Hif1 α is a key mediator of cellular adjustment to nutrient and oxygen stress [53], acting as a transcriptional activator of multiple genes, inhibiting mitochondrial respiration by up-regulating expression of the pyruvate dehydrogenase kinase (PDK) gene [54,55]. Under normoxia, Hif1 α is hydroxylated at multiple prolyl residues by the prolyl-hydroxylase-domain (PHD) proteins, adjusting basal expression of its target genes, being tightly regulated under normal nutrient conditions. But, in conditions of low levels of oxygen or glucose, PHD protein is inactivated, stabilizing Hif1 α protein level [56]. Zhong and collaborates [57] demonstrated that SIRT6 deficiency leads to a upregulation of glucose uptake, triggering a nutrient-stress response and a switch of glucose metabolism towards glycolysis rather than mitochondrial respiration, repressing the Hif1 α transcriptional activity and deacetylating H3K9. But, under normal conditions, SIRT6 maintains glucose flux into the Krebs cycle and oxidative phosphorylation.

Liu and coworkers [58] demonstrated that satellite cells cultured under hypoxic condition have diminished differentiation capacity but improved self-renewal potential, leading to increased transplantation efficiency, through upregulating Pax7 and Notch signaling pathways, mediated by downregulating miR-1 and miR-206 [58]. Similarly, Cerletti et al. [59] showed that the effectiveness of satellite cell transplantation is improved by calorie restriction. Additionally, this condition significantly enhanced myogenic function and cell proliferative capacity, associated with Pax7⁺/myofiber content. These alterations were correlated to NAD⁺-dependent protein deacetylase SIRT1, increased oxygen consumption rate and mitochondrial function, and reduced lactate production.

The mTOR pathway has been suggested to regulate the function and stem cell fate decisions [60,61]. Rodgers and coworkers [62] showed that stem cell quiescent state is constituted of two divergent functional stages, G₀ and “alert” state or G_{Alert}, this last one is characterized by increased cell size and transcriptional activity, mitochondrial metabolism and accelerated cell cycle entry. The authors demonstrated that mTORC1 activated *via* Akt/PI3K pathway signaling is required for the passage of satellite cells from G₀ into G_{Alert}. In addition, the transition G₀ to G_{Alert} improves the muscle regenerative process by increasing myogenin expression and cell fusion. Other quiescent cells, such as fibro-adipogenic progenitors, resident mesenchymal stem cells and hematopoietic stem cells, respond in a similar way [62–64].

In vivo, studies with endurance exercise are associated with increased fatty acid oxidation, mitochondrial deacetylation in rest, and during exercise, suggesting a posttranslational modification modulated

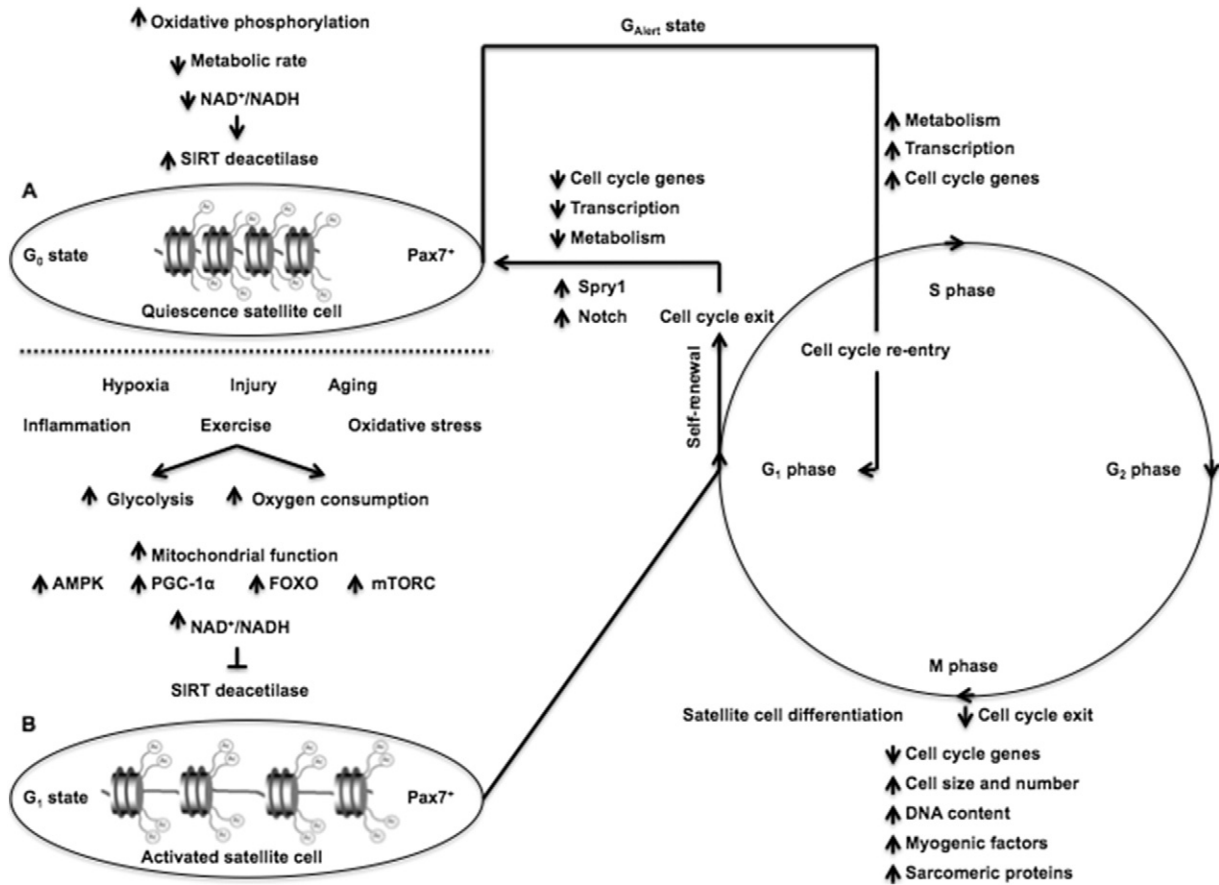


Fig. 1. The transition from quiescence muscle satellite cells to activation involves a metabolic switch from oxidative phosphorylation to glycolytic metabolism. Satellite cell quiescence displays a dormant state known as G₀ state, being constituted of two different functional phases, G₀ and an “alert” phase we term G_{Alert}. In this condition, quiescent satellite cells have a low metabolic rate (A). The activation and entry into the cell cycle are characterized by change from oxidative to glycolytic metabolism. Following activation, satellite cells can exit the cell cycle and return to quiescence. In response to some conditions, such as increased the expression of exercise-responsive genes in skeletal muscle, enhanced exercise performance, increased whole-body energy expenditure and fatty acid oxidation, a metabolic switch not yet known, drives the satellite cell activation that results in improved differentiation and/or self-renewal. This metabolic alteration produces ATP while concurrently decreased the cellular level of NAD⁺ and nuclear SIRT1 deacetylase (B). The loss or defects of satellite cell activity are correlated with an extensive diversity of muscular illnesses. The preservation of muscle mass and regenerative potential depends on the permanent existence of a functional pool of self-renewing muscle satellite cells, highlighting the significant importance of metabolic activity influenced by exercise on muscle physiology. Remarkably, newly findings suggest a metabolic control of satellite cells activation. However, *in vivo* studies into the molecular proceedings following metabolic alterations in satellite cells are indispensable and will likely lead to the identification of new cellular targets that control satellite cell activation and myonuclei accretion in adult skeletal muscle.

by aerobic muscle adaptation and increased of maximal oxygen consumption (VO₂max) [74]. This approach could be a helpful preventive intervention to modulate metabolic pathways in chronic disorders. Together, various cellular targets and metabolic factors governing satellite cell functions have been described. These findings suggest that the metabolic status of satellite cells has an important role in the control of cell myogenic commitment.

4. Satellite cell activation in response to aerobic exercise adaptation: evidence in rodent models

Adaptations to endurance exercise include repeated transient changes in gene expression that modify biomolecule synthesis, quantities, and, ultimately, phenotype. These adaptations involve the expression of exercise-responsive proteins in skeletal muscle related to exercise performance, mitochondrial biogenesis, and oxidative phosphorylation capacity, as well as whole-body energy expenditure and fatty acid oxidation. It has been demonstrated that these processes are highly complex and involve a well-defined and controlled network of transcription factors and coregulator proteins regulating gene expression profile during endurance exercise training [6,16,17,65–71].

Recently, some findings suggest that endurance exercise induces increase in muscle satellite cell amount. But the mechanisms underlying this effect have not been investigated yet. Evidence suggests that it

can be associated with a dramatic cellular metabolic reprogramming, favoring oxidative rather than glycolytic metabolism, leading to pivotal adjustments for mitochondrial mass and function. Understanding the mechanisms mediating exercise adaptations could provide new insights for the discovery of potential molecular targets for preventive and/or therapeutic intervention of metabolic diseases [72–74].

During sessions of resistance or workload exercise, satellite cells are activated and contribute to myonuclear expansion and hypertrophy. However, the mechanisms involved in the function of satellite cell activation in response to endurance exercise remains to be investigated yet [2,3,5]. For evaluation of satellite cell contribution to muscle plasticity after different conditions, including different physical exercise protocols, genetic mouse models have been developed. The conditional ablation of Pax7-expressing satellite cells in mature skeletal muscle results in a complete absence of regeneration [2–4], but adult muscle hypertrophy [3] and regrowth subsequent atrophy [75] were not modified, suggesting that satellite cells have differential functions during muscle repair, hypertrophy, and regrowth.

Recently, studies with endurance exercise protocols have shown a significant increased amount of satellite cells and myofibers. In aged rats, the aerobic adaptation induced by endurance exercise stimulates the myogenesis, without changing the fiber area or muscle hypertrophy, suggesting that aerobic exercise also enhances the satellite cell number and myogenesis in aged rats [6,15].

In rodents, the increased aerobic muscle adaptation induced by spontaneous activity in wheel free running or treadmill exercise training has been related with alteration in fiber type composition and satellite cell expansion, appointing out the fuel selection and mitochondrial oxidative efficiency as mechanisms linking enhanced aerobic exercise capacity [6,8,9,15,71,76–80] (Table 1).

Satellite cell number is increased, but addition of myonuclei is not, during post-natal growth (4–10 weeks of age) in response to endurance exercise, suggesting that cell proliferation and fusion are not necessarily linked. Interestingly, the high-load aerobic exercise (resistance required to move the wheel running was progressively increased every 4 days, totalizing an elevation of 173%, over the 6 weeks) enhances the satellite cell number and myonuclei in several skeletal muscles [72], suggesting that the load of muscle activity is essential for improving satellite cell amount and its conversion to myonuclei.

In the rat *plantaris* muscle, a correlation between satellite cell quantity and wheel-running performance (increased oxidative capacity) has been observed, suggesting that satellite cell content is directly proportional to the distance run. The trained group (characterized by low-intensity and long-duration exercise training), for eight weeks, showed no differences in muscle weight or fiber area (suggesting absence of hypertrophy), but the amount of satellite cells and myonuclei were significantly elevated [15].

Jackson et al. [76], using the Pax7/DTA mouse model, which presents severe depletion of satellite cells (~90%) in adult skeletal muscle, investigated the participation of satellite cells on aerobic muscle adaptation induced by wheel running exercise, during eight weeks. The authors found reduction of wheel running performance, but no disturbing markers for aerobic adaptation, such as mitochondrial content, capillary density, and fiber-type composition. Intriguingly, the authors demonstrated that muscle spindles and intrafusal fiber size are negatively impacted by Pax7 ablation, apart from ECM deposition, suggesting an important role of satellite cells on this process [78,81].

It has been observed that aerobic exercise alleviates the progressive loss of skeletal muscle mass and muscle strength aging-induced in mice [82,83]. In this context, Cisterna and collaborators [8] demonstrated that aerobic exercise increases the satellite cell amount and activation, as well as differentiation into myotubes. Effect of aerobic training on decline of skeletal muscle mass and performance, and reduction of satellite cells in fast-twitch myofibers during aging in male and female rats, was analyzed in a similar study [6]. The authors showed that running exercise induces a significant increase in the abundance of myofibers containing of elevated number of satellite cells [6].

Other rodent studies using male Sprague–Dawley rats also appoint out for the effects of aerobic training on satellite cell physiology. These animals were housed in resistance wheel running (with increasing torque required to move the wheel, reaching the load of 173% per body weight) or in conventional free-spinning running wheel, for 45 days. Animals submitted to the resistance wheel running, compared to the conventional free-spinning wheel, improved the wet muscle

mass and fiber cross sectional area of the *plantaris*, *vastus lateralis*, *extensor carpi radialis longus*, *brevis*, and *soleus* muscles, suggesting that the kind of voluntary movement is important for the physiological muscle adaptation to aerobic exercise [78]. Similar results in *plantaris* muscle mass was also demonstrated following eight weeks of endurance exercise, with a maximum load of 74% total body mass [84]. Moreover, controlling a progressive wheel loading protocol in adult mice (41% per body weight), there is not alteration in the *plantaris*, *gastrocnemius* or *tibialis anterior* muscle mass; however, authors found an increase of the soleus muscle weight by approximately 20% following seven weeks of endurance exercise [84,85]. In response to resistance wheel running exercise (increasing resistance over 10 weeks), besides of improved muscle aerobic capacity, Soffe et al. [9] observed an elevated dry mass (hypertrophy) of the *soleus*, *quadriceps*, and *gastrocnemius* muscles of male C57Bl/6J mice, associated with elevated muscle cross sectional area.

Fujimaki and coworkers [81] examined the properties of Wnt/ β -catenin signaling on the regulation of satellite cells following aerobic voluntary wheel running for four weeks in adult and aged mice. Interestingly, there were no changes in body mass or wet weight, besides Akt and p70^{S6K} phosphorylation was altered. However, the study indicated that the number of Pax7⁺, Pax7⁺ MyoD⁺, and Pax7⁺ Myf5⁺ cells was increased, as well as, the number of BrdU⁺ cells (proliferating cells) in *gastrocnemius* muscle following wheel running in both adult and aged mice. Additionally, the expression of Delta ligand 4 (DLL4) and HeyL, ligand and target gene of Notch signaling, respectively, mildly decreases following voluntary exercise in both the adult and aged groups. However, the expression of Wnt3, Wnt5a, and Wnt5b ligands is expressively augmented following voluntary wheel running exercise, being this up-regulation observed in both groups, adult and aged groups. In this context, Wnt signaling seems to play an essential function in further stimulating myogenic differentiation in lineage-committed progenitor cells, without significant effect on undifferentiated satellite cells, in which Notch signaling regulates this process under normal physiological conditions [86]. Furthermore, the phosphorylation of GSK-3 (Ser-9) slightly rises and the expression of β -catenin is highly augmented following running exercise, suggesting that canonical Wnt/ β -catenin signaling is stimulated during the exercise session. Based on these results, Fujimaki and collaborators [81] hypothesized that voluntary wheel running stimulates the activation of satellite cells, transcription of Myf5⁺ and MyoD⁺ (myogenesis), expression of canonical Wnt member proteins, activity of canonical Wnt/ β -catenin signaling, contributing to the specific gene expression profile in both adult and aged skeletal muscle, without producing fibrosis, even in aged skeletal muscle.

5. Satellite cell activation in response to aerobic exercise adaptation: evidence in human

Remarkably, numerous studies have revealed that the muscle mass is increased following resistance exercise (muscle hypertrophic

Table 1
Satellite cells activation in response to aerobic exercise adaptation in rodents.

Time of measure (acute/chronic)	Exercise type	Specie (sex/age)	Muscle analyzed	Marker used	Satellite cell number	Myonuclei number	Exercise performance	Fiber diameter	Reference
Chronic	FW	Mice – Pax7-DTA (female/6 months) ^a	PM	Pax7	↔	↔	↑ ^a	↔	Jackson et al. [76]
Chronic	FW, RW	Rats (male/4, 5, 7 or 10 weeks)	GM, Sol, VL,TA	Pax7	↑	↑		↑	Smith and Merry, [72]
Chronic	TM	Rats (male/15–17 months) (female/15 months) young – 3.5 months	G	Pax7	↑		↑		Shefer et al. [6]
Chronic	FW	Rats (male/5 weeks)	PM	Pax7	↑	↑	↑	↔	Kurosaka et al. [15]
Chronic	FW	Mice (male/8–12 weeks and 24 months)	GM	Pax7	↑				Fujimaki et al. [81]
Chronic	Endurance	Rats (female/17 weeks)	PM	Pax7	↑	↔		↔	Kurosaka et al. [70]
Chronic/acute	TM	Rats (female/10–12 weeks)	Sol, TA, VI	BrdU	↑	↑		↔	Smith et al. [87]
Chronic	TM	Mice (male/12–28 months)	Q	Pax7	↑	↑			Cisterna et al. [8]

TM – Treadmill-running; EDL – *Extensor digitorum longus*; GM – *Gastrocnemius muscle*; VI – *Vastus intermedius*; TA – *Tibialis anterior*; PM – *Plantaris muscle*; Sol – *Soleus*; FW – Voluntary free wheel running; RW – Voluntary resistance wheel running; VL – *Vastus lateralis*; Q – *Quadriceps*.

^a Vehicle-treated Pax7/DTA mice.

stimulus) in humans. This effect has been associated with myonuclei accretion and expansion of satellite cells [10,11,14]. In this context, considerable importance has been addressed on describing the function of satellite cells in response to resistance exercise, but little emphasis is directed on examining the function of satellite cells in response to aerobic exercise adaptation.

Some studies reported an increase of the satellite cell number and muscle cross-sectional area of fiber type IIa following intense aerobic intermittent cycle ergometer training [15,16], whereas others reported no alteration using a low-intensity (moderate intensity) aerobic exercise based on the VO_2max [19] (Table 2). These results suggest that intensity of aerobic exercise, rather than duration, plays a fundamental function in expansion of satellite cells. In this context, if satellite cells play a role in response to aerobic exercise adaptation, which one leads to muscle stress without significant muscle hypertrophy, has not been addressed yet.

Charifi et al. [16] investigated the modulation of satellite cells and myonuclear expansion induced by aerobic muscle adaptation following endurance exercise for 14 weeks, in bicycle ergometer (4 min at 65–75% of peak of VO_2max , followed by 1 min at 85–95% VO_2max , during 30 min). The endurance exercise resulted in increased VO_2max and citrate synthase activity. The content of satellite cell and type IIa fiber were increased, while there was no statistical difference in myonuclear number and fiber size (hypertrophy) [16], demonstrating that satellite cell expansion in response to endurance exercise can represent a new signaling target for therapeutic intervention during aging.

In other study, Verney et al. [17] described the expansion of satellite cell pool associated with type II fibers in older adults, following endurance exercise for 14 weeks. There were no significant changes in myonuclear number or myonuclear domain after exercise [17]. In contrast, endurance exercise during six months does not lead to alteration in the satellite cell amount in obesity-related type 2 diabetic patients [18].

A recent and intriguing work that investigated the effect of endurance exercise on satellite cell modulation, Fry et al. [7] demonstrated that a protocol of twelve weeks of aerobic exercise, performed on cycle ergometer twice a week, by 45 min at 70% of the heart rate reserve, increases the cross-sectional area in both MyHC type I and type IIa fibers, related to expansion of the satellite cell pool and myonuclear number specifically in MyHC type I. Additionally, hypertrophy of MyHC type II fibers occurred without detectable myonuclear accretion or changes in the satellite cell number, suggesting for a fiber type-specific role of satellite cells on aerobic muscle adaptation following endurance exercise.

More recently, Nederveen et al. [88] examined the acute response of satellite cell content, 24 h and 48 h following high-intensity or moderate-intensity interval endurance exercise, in twenty-two untrained healthy old adult men. There was a significant increase in the satellite cell content, as demonstrated by elevated expressing Pax7⁺ cells associated exclusively with type I fibers. Additionally, satellite cells expressing

MyoD⁺ (a marker for satellite cell myogenesis) was elevated 48 h after both intensity exercise protocol.

Interestingly, aerobic fitness and endurance capacity in health subjects were measured by VO_2max level and correlated to the muscle satellite cell expansion. A positive relationship was found between the amount of Pax7⁺ cells and the VO_2max . This finding suggests that an aerobic physically active lifestyle able to improve VO_2max , stimulates the satellite cell expansion [89].

Joanisse et al. [18] investigated the function of satellite cells in response to high-intensity endurance exercise based in VO_2max , during 6 weeks, and examined whether the increase of satellite cells is associated with remodeling muscle fibers. There was no substantial increase in muscle cross sectional area for either type I, II, or hybrid fibers. Also, there was no modification of the number of nuclei per fiber, as well as the myonuclear domain or hypertrophic stimuli following exercise. Additionally, the exercise reduced type II fibers, while the content of type I fibers was unaffected. Increased satellite cell content was observed in association with hybrid fibers (fibers expressing myosin heavy-chain type I and II isoforms, concomitantly). Furthermore, the authors observed that hybrid fibers were positively correlated with Pax7⁺/MyoD⁺ quiescent satellite cells and Pax7⁺/MyoD⁺ differentiated cells following exercise [18]. Later, the same group investigated the effect of high-, moderate-, and low-intensity cycling endurance exercise, during six weeks, on fiber type-specific satellite cell expansion in untrained men and women. The different intensities of exercise did not induce hypertrophy. However, there was an increase in the activated satellite cells (Pax7⁺/MyoD⁺ and Pax7⁺/MyoD⁺) after each different exercise, possibly contributing to exercise-induced muscle aerobic adaptations [90].

Mackey et al. [91] observed that old men, when submitted to aerobic exercise, have a high content of type I fibers and satellite cell content compared with young untrained or old untrained healthy men. Overall, there was a positive association between satellite cell expansion and type I fibers in exercised humans, independently of age. These results suggest that increased exercise-responsive gene expression in skeletal muscle in response to aerobic exercise stimulates the accretion of satellite cells [91].

Taken together, evidence from human studies strongly suggests that satellite cells play a key role in the muscle aerobic metabolic adaptation. Thus, while the requirement of muscle satellite cells and myonuclear accretion following muscle injury or resistance exercise (hypertrophic stimulus) has been demonstrated, the mechanisms involved during skeletal muscle aerobic metabolic adaptations, including enhanced functional capacity and mitochondrial function in response to endurance exercise is not completely understood yet (Fig. 2).

6. Conclusion

Skeletal muscle is a highly dynamic and metabolically active tissue that can self-repair in response to injury by recruiting muscle stem cells, known as satellite cells. After activation, these cells can expand,

Table 2
Satellite cells activation in response to aerobic exercise adaptation in humans.

Time of measure (acute/chronic)	Exercise type	Specie (sex/age)	Muscle analyzed	Marker used	Satellite cell content (fiber type)	Myonuclear content (fiber type)	Exercise performance	Fiber diameter	Reference
Chronic	Aerobic	Human (men/70–80)	VL	CD56	↑	↔	↑	↑ IIa	Charifi et al. [16]
Chronic	Aerobic	Human (men/73 ± 4)	VL	CD56	↑	↔	↑	↑	Verney et al. [17]
Chronic	Aerobic	Human (obese/diabetic men/61 ± 6)	VL	CD56	↔	↔	↔	↔	Snijders et al. [19]
Chronic	Aerobic	Human (women/men/27 ± 8)	VL	Pax7	↔	↔	?	↔	Joanisse et al. [18]
Chronic	Aerobic	Human (men/43 ± 16)	VL	Pax7	↑	↑	↑	↑	Mackey et al. [91]
Chronic	Aerobic	Human (women/men/47 ± 6)	VL	Pax7	↑ (I)	↑ (I)	↑	↑	Fry et al. [7]
Acute	Resistance/aerobic	Human (men/67 ± 4)	VL	Pax7	↑	↑	↑	↑	Nederveen et al. [88]
Chronic	Aerobic	Human (women/men/21 ± 4)	VL	Pax7	↑	↔	?	↔	Joanisse et al. [90]
Acute	Aerobic	Human (men/24 ± 3)	VL	Pax7	↑	↑	↑	↑	Macaluso et al. [89]

T - Muscle trapezius; VL - Muscle vastus lateralis; D - Muscle deltoid.

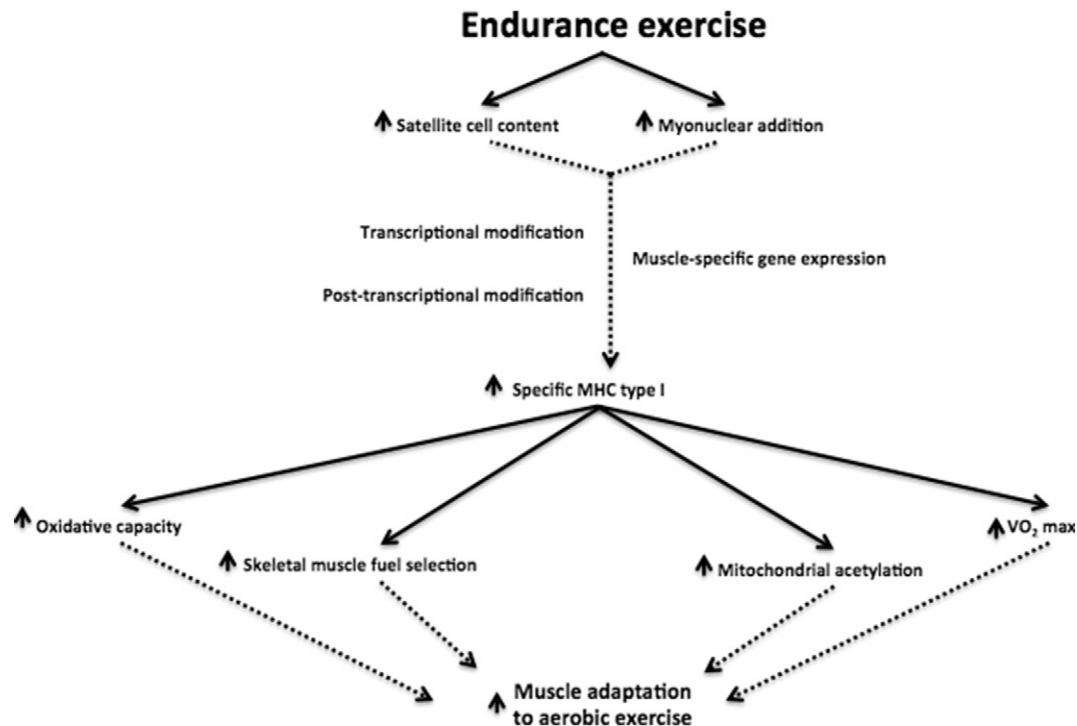


Fig. 2. Fiber type-specific satellite cell expansion induced by mitochondrial oxidative efficiency and muscle aerobic adaptation in response to endurance exercise. It's has been well established that satellite cell expansion is accompanied by hypertrophy stimulus induced by resistance exercise, but currently their mechanisms on enhanced skeletal muscle aerobic adaptations, including the growth of myosin heavy chain (MHC) type I, maximal oxygen consumption (VO_2max) and mitochondrial content and function in response to endurance exercise remains not known. Future studies will needed to investigate the molecular responses to endurance exercise to validate the effective targets to pharmacologically manipulate aspects of the exercise adaptive response. This approach could be a useful prevention strategy to enhance muscle metabolic health in chronic diseases.

contributing for the muscle aerobic adaptation in response to endurance exercise. The trigger for this activation and expansion process is under investigation, but it seems to be a multifaceted and a well-controlled response. This review presented herein discusses how endurance exercise alters the physiology and function of muscle satellite cells with significant beneficial effects in enhancing myogenic activity. *In vitro* studies showed that inactivated satellite cells have low metabolic rate, but when activated, these cells entry into the cell cycle, with markedly major metabolic changes, indicating a possible function for metabolic control and oxygen availability in satellite cell fate control, enhancing their self-renewal potential. These results support the idea that multiple cell factors, especially metabolic regulators, contribute to muscle regenerative function in response to aerobic exercise adaptation and, therefore, they can be potential target candidates for therapeutic intervention for muscle metabolic disorders.

Conflicts of interest statement

The authors declare no conflict of interest.

Acknowledgments

This research was supported by grants from FAPESP (2016/18633-8 and 2011/06815-0), CNPq (457358/2013-0, 407849/2013-0 e 472090/2013-5), CAPES (88887.124164/2014-00), National Institute of Sciences and Technology in Obesity and Diabetes (INCT 2009), Dean's Office for Research (NAP 2013), University of Sao Paulo, Dean's Office for Post-graduate Studies and Research of the Cruzeiro do Sul University (2014/33078017), and Guggenheim Foundation (04/2008).

References

- [1] A. Mauro, Satellite cell of skeletal muscle fibers, *J. Biophys. Biochem. Cytol.* 9 (1961) 493–495.

- [2] R. Sambasivan, R. Yao, A. Kissenpfennig, L. Van Wittenberghe, A. Paldi, B. Gayraud-Morel, H.B. Guenou, S. Malissen, A.G. Tajbakhsh, Pax7-expressing satellite cells are indispensable for adult skeletal muscle regeneration, *Development* 138 (2011) 3647–3656.
- [3] J.J. McCarthy, J. Mula, M. Miyazaki, R. Erfani, K. Garrison, A.B. Farooqui, R. Srikuea, B.A. Lawson, B. Crimes, C. Keller, G. Van Zant, K.S. Campbell, K.A. Esser, E.E. Dupont-Versteegden, C.A. Peterson, Effective fiber hypertrophy in satellite cell-depleted skeletal muscle, *Development* 138 (2011) 3657–3666.
- [4] C. Lepper, T.A. Partridge, C.M. Fan, An absolute requirement for Pax7-positive satellite cells in acute injury-induced skeletal muscle regeneration, *Development* 138 (2011) 3639–3646.
- [5] M.M. Murphy, J.A. Lawson, S.J. Mathew, D.A. Hutcheson, G. Kardon, Satellite cells, connective tissue fibroblasts and their interactions are crucial for muscle regeneration, *Development* 138 (2011) 3625–3637.
- [6] G. Shefer, G. Rauner, Z. Yablonska-Reuveni, D. Benayahu, Reduced satellite cell numbers and myogenic capacity in aging can be alleviated by endurance exercise, *PLoS One* 5 (2010), e13307.
- [7] C.S. Fry, B. Noehren, J. Mula, M.F. Ubele, P.M. Westgate, P.A. Kern, C.A. Peterson, Fibre type-specific satellite cell response to aerobic training in sedentary adults, *J. Physiol.* 592 (2014) 2625–2635.
- [8] B. Cisterna, M. Giagnacovo, M. Costanzo, P. Fattoretti, C. Zancanaro, C. Pellicciari, M. Malatesta, Adapted physical exercise enhances activation and differentiation potential of satellite cells in the skeletal muscle of old mice, *J. Anat.* 228 (5) (2016) 771–783.
- [9] R. Soffe, S. Baratchi, S.Y. Tang, A. Mitchell, P. McIntyre, K. Khoshmanesh, Concurrent shear stress and chemical stimulation of mechano-sensitive cells by discontinuous dielectrophoresis, *Biomicrofluidics* 10 (2) (2016) 024117.
- [10] F. Kadi, L.E. Thornell, Concomitant increases in myonuclear and satellite cell content in female trapezius muscle following strength training, *Histochem. Cell Biol.* 113 (2000) 99–103.
- [11] S.M. Roth, G.F. Martel, F.M. Ivey, J.T. Lemmer, B.L. Tracy, E.J. Metter, B.F. Hurley, M.A. Rogers, Skeletal muscle satellite cell characteristics in young and older men and women after heavy resistance strength training, *J. Gerontol. A Biol. Sci. Med. Sci.* 56 (6) (2001) B240–B247.
- [12] J.K. Petrella, J.S. Kim, J.M. Cross, D.J. Kosek, M.M. Bamman, Efficacy of myonuclear addition may explain differential myofiber growth among resistance-trained young and older men and women, *Am. J. Physiol. Endocrinol. Metab.* 291 (2006) E937–E946.
- [13] J.K. Petrella, J.S. Kim, D.L. Mayhew, J.M. Cross, M.M. Bamman, Potent myofiber hypertrophy during resistance training in humans is associated with satellite cell-mediated myonuclear addition: a cluster analysis, *J. Appl. Physiol.* 104 (2008) 1736–1742.
- [14] L.B. Verdijk, B.G. Gleeson, R.A.M. Jonkers, K. Meijer, H. Savelberg, P. Dendale, L.J.C. van Loon, Skeletal muscle hypertrophy following resistance training is accompanied

- by a fiber type-specific increase in satellite cell content in elderly men, *J. Gerontol. A Biol. Sci. Med. Sci.* 64 (2009) 332–339.
- [15] M. Kurosaka, H. Naito, Y. Ogura, A. Kojima, K. Goto, S. Katamoto, Effects of voluntary wheel running on satellite cells in the rat plantaris muscle, *J. Sports Sci. Med.* 8 (2009) 51–57.
 - [16] N. Charifi, F. Kadi, L. Féasson, C. Denis, Effects of endurance training on satellite cell frequency in skeletal muscle of old men, *Muscle Nerve* 28 (1) (2003) 87–92.
 - [17] J. Verney, F. Kadi, N. Charifi, L. Féasson, M.A. Saafi, J. Castells, K. Piehl-Aulin, C. Denis, Effects of combined lower body endurance and upper body resistance training on the satellite cell pool in elderly subjects, *Muscle Nerve* 38 (3) (2008) 1147–1154.
 - [18] S. Joannis, J.B. Gillen, L.M. Bellamy, B.R. McKay, M.A. Tarnopolsky, M.J. Gibala, G. Parise, Evidence for the contribution of muscle stem cells to nonhypertrophic skeletal muscle remodeling in humans, *FASEB J.* 27 (11) (2013) 4596–4605.
 - [19] T. Snijders, L.B. Verdijk, D. Hansen, P. Dendale, L.J. van Loon, Continuous endurance-type exercise training does not modulate satellite cell content in obese type 2 diabetic patients, *Muscle Nerve* 43 (3) (2011) 393–401.
 - [20] J.G. Ryall, S. Dell'Orso, A. Derfoul, A. Juan, H. Zare, X. Feng, D. Clermont, M. Koulis, G. Gutierrez-Cruz, M. Fulco, V. Sartorelli, The NAD(+)–dependent SIRT1 deacetylase translates a metabolic switch into regulatory epigenetics in skeletal muscle stem cells, *Cell Stem Cell* 16 (2) (2015) 171–183.
 - [21] T.J. Hawke, D.J. Garry, Myogenic satellite cells: physiology to molecular biology, *J. Appl. Physiol.* 91 (2001) 534–551.
 - [22] R.B. White, A.S. Biérinx, V.F. Gnocchi, P.S. Zammit, et al., *BMC Dev. Biol.* 10 (2010) 21.
 - [23] F. Relaix, P.S. Zammit, Satellite cells are essential for skeletal muscle regeneration: the cell on the edge returns centre stage, *Development* 139 (16) (2012) 2845–2856.
 - [24] X. Shi, D. Garry, Muscle stem cells in development, regeneration and disease, *Genes Dev.* 20 (2006) 1692–1708.
 - [25] P. Seale, L.A. Sabourin, A. Girgis-Gabardo, A. Mansouri, P. Gruss, M.A. Rudnicki, Pax7 is required for the specification of myogenic satellite cells, *Cell* 102 (2000) 777–786.
 - [26] L.K. McLoon, J. Wirtschafter, Activated satellite cells in extraocular muscles of normal adult monkeys and humans, *Invest. Ophthalmol. Vis. Sci.* 44 (2003) 1927–1932.
 - [27] C.L. Hammond, Y. Hinitz, D.P. Osborn, J.E. Minchin, G. Tettamanti, S.M. Hughes, Signals and myogenic regulatory factors restrict pax3 and pax7 expression to dermomyotome-like tissue in zebrafish, *Dev. Biol.* 302 (2007) 504–521.
 - [28] O. Halevy, Y. Piestun, M.Z. Allouh, B.W. Rosser, Y. Rinkevich, R. Reshef, I. Rozenboim, M. Wlekinski-Lee, Z. Yablonska-Reuveni, Pattern of Pax7 expression during myogenesis in the posthatch chicken establishes a model for satellite cell differentiation and renewal, *Dev. Dyn.* 231 (2004) 489–502.
 - [29] Y. Chen, J.D. Zajac, H.E. MacLean, Androgen regulation of satellite cell function, *J. Endocrinol.* 186 (2005) 21–31.
 - [30] M. Patrino, F. Caliaro, T. Martinello, F. Mascarello, Expression of the paired box domain Pax7 protein in myogenic cells isolated from the porcine semitendinosus muscle after birth, *Tissue Cell* 40 (2008) 1–6.
 - [31] J.I. Morrison, S. Loof, P. He, A. Simon, Salamander limb regeneration involves the activation of a multipotent skeletal muscle satellite cell population, *J. Cell Biol.* 172 (2006) 433–440.
 - [32] E. Schultz, M.C. Gibson, T. Champion, Satellite cells are mitotically quiescent in mature mouse muscle: an EM and radioautographic study, *J. Exp. Zool.* 206 (1978) 451–456.
 - [33] M. Buckingham, L. Bajard, T. Chang, P. Daubas, J. Hadchouel, S. Meilhac, D. Montarras, D. Rocancourt, F. Relaix, The formation of skeletal muscle: from somite to limb, *J. Anat.* 202 (2003) 59–68.
 - [34] D.D. Cornelison, B.J. Wold, Single-cell analysis of regulatory gene expression in quiescent and activated mouse skeletal muscle satellite cells, *Dev. Biol.* 191 (1997) 270–283.
 - [35] A. Irintchev, M. Zeschig, A. Starzinski-Powitz, A. Wernig, Expression pattern of Mcadherin in normal, denervated, and regenerating mouse muscles, *Dev. Dyn.* 199 (1994) 326–337.
 - [36] J.R. Beauchamp, L. Heslop, D.S. Yu, S. Tajbakhsh, R.G. Kelly, A. Wernig, M.E. Buckingham, T.A. Partridge, P.S. Zammit, Expression of CD34 and Myf5 defines the majority of quiescent adult skeletal muscle satellite cells, *J. Cell Biol.* 151 (2000) 1221–1234.
 - [37] R.E. Allen, S.M. Sheehan, R.G. Taylor, T.L. Kendall, G.M. Rice, Hepatocyte growth factor activates quiescent skeletal muscle satellite cells in vitro, *J. Cell. Physiol.* 165 (1995) 307–312.
 - [38] V.F. Gnocchi, R.B. White, Y. Ono, J.A. Ellis, P.S. Zammit, Further characterisation of the molecular signature of quiescent and activated mouse muscle satellite cells, *PLoS One* 4 (2009), e5205.
 - [39] D.D. Cornelison, M.S. Filla, H.M. Stanley, A.C. Rapraeger, B.B. Olwin, Syndecan-3 and syndecan-4 specifically mark skeletal muscle satellite cells and are implicated in satellite cell maintenance and muscle regeneration, *Dev. Biol.* 239 (2001) 79–94.
 - [40] M.Z. Ratajczak, M. Majka, M. Kucia, J. Drukala, Z. Pietrzkowski, S. Peiper, A. Janowska-Wieczorek, Expression of functional CXCR4 by muscle satellite cells and secretion of SDF-1 by muscle-derived fibroblasts is associated with the presence of both muscle progenitors in bone marrow and hematopoietic stem/progenitor cells in muscles, *Stem Cells* 21 (2003) 363–371.
 - [41] S. Fukada, A. Uezumi, M. Ikemoto, S. Masuda, M. Segawa, N. Tanimura, H. Yamamoto, Y. Miyagoe-Suzuki, S. Takeda, Molecular signature of quiescent satellite cells in adult skeletal muscle, *Stem Cells* 25 (2007) 2448–2459.
 - [42] R. Meech, K.N. Gonzalez, M. Barro, A. Gromova, L. Zhuang, J.A. Hulin, H.P. Makarenko, Barx2 is expressed in satellite cells and is required for normal muscle growth and regeneration, *Stem Cells* 30 (2012) 253–265.
 - [43] H. Schmalbruch, U. Hellhammer, The number of nuclei in adult rat muscles with special reference to satellite cells, *Anat. Rec.* 189 (2) (1977) 169–175.
 - [44] M.C. Gibson, E. Schultz, Age-related differences in absolute numbers of skeletal muscle satellite cells, *Muscle Nerve* 6 (8) (1983) 574–580.
 - [45] S.Y. Lunt, M.G. Vander Heiden, Aerobic glycolysis: meeting the metabolic requirements of cell proliferation, *Annu. Rev. Cell Dev. Biol.* 27 (2011) 441–464.
 - [46] J.G. Ryall, Metabolic reprogramming as a novel regulator of skeletal muscle development and regeneration, *FEBS J.* 280 (2013) 4004–4013.
 - [47] N. Shyh-Chang, G.Q. Daley, L.C. Cantley, Stem cell metabolism in tissue development and aging, *Development* 140 (2013) 2535–2547.
 - [48] S. Katada, A. Imhof, P. Sassone-Corsi, Connecting threads: epigenetics and metabolism, *Cell* 148 (2012) 24–28.
 - [49] C. Lu, C.B. Thompson, Metabolic regulation of epigenetics, *Cell Metab.* 16 (2012) 9–17.
 - [50] J. Segalés, E. Perdiguero, P. Muñoz-Cánoves, *FEBS J.* 22 (September 2014) Published online.
 - [51] M.V. Gustafsson, X. Zheng, T. Pereira, K. Gradin, S. Jin, J. Lundkvist, J.L. Ruas, L. Poellinger, U. Lendahl, M. Bondesson, et al., *Dev. Cell* 9 (2005) 617–628.
 - [52] A.J. Majumdar, N. Skuli, R.C. Mesquita, M.N. Kim, A.G. Yodh, M. Nguyen-McCartayand, M.C. Simon, O₂ regulates skeletal muscle progenitor differentiation through PI3K/AKT signaling, *Mol. Cell. Biol.* 32 (2012) 36–49.
 - [53] J.J. Lum, T. Bui, M. Gruber, J.D. Gordon, R.J. DeBerardinis, K.L. Covello, M.C. Simon, C.B. Thompson, The transcription factor HIF-1 α plays a critical role in the growth factor-dependent regulation of both aerobic and anaerobic glycolysis, *Genes Dev.* 21 (2007) 1037–1049.
 - [54] J.W. Kim, I. Tchernyshyov, G.L. Semenza, C.V. Dang, HIF-1-mediated expression of pyruvate dehydrogenase kinase: a metabolic switch required for cellular adaptation to hypoxia, *Cell Metab.* 3 (2006) 177–185.
 - [55] I. Papandreou, R.A. Cairns, L. Fontana, A.L. Lim, N.C. Denko, HIF-1 mediates adaptation to hypoxia by actively downregulating mitochondrial oxygen consumption, *Cell Metab.* 3 (2006) 187–197.
 - [56] J. Aragonés, P. Fraisl, M. Baes, P. Carmeliet, Oxygen sensors at the crossroad of metabolism, *Cell Metab.* 9 (1) (2009) 11–22.
 - [57] L. Zhong, A. D'Urso, D. Toiber, C. Sebastian, R.E. Henry, D.D. Vadysirisack, A. Guimaraes, B. Marinelli, J.D. Wikstrom, T. Nir, C.B. Clish, et al., The histone deacetylase Sirt6 regulates glucose homeostasis via Hif1 α , *Cell* 140 (2) (2010) 280–293.
 - [58] W. Liu, Y. Wen, P. Bi, X. Lai, X.S. Liu, X. Liu, S. Kuang, Hypoxia promotes satellite cell self-renewal and enhances the efficiency of myoblast transplantation, *Development* 139 (16) (2012) 2857–2865.
 - [59] M. Cerletti, Y.C. Jang, L.W. Finley, M.C. Haigis, A.J. Wagers, Short-term calorie restriction enhances skeletal muscle stem cell function, *Cell Stem Cell* 10 (5) (2012) 515–519.
 - [60] J. Zhou, P. Su, L. Wang, J. Chen, M. Zimmermann, O. Genbacev, O. Afonja, M.C. Horne, T. Tanaka, E. Duan, S.J. Fisher, J. Liao, F. Wang, mTOR supports long-term self-renewal and suppresses mesoderm and endoderm activities of human embryonic stem cells, *Proc. Natl. Acad. Sci. U. S. A.* 106 (2009) 7840–7845.
 - [61] C.A. Easley, A. Ben-Yehudah, C.J. Redinger, S.L. Oliver, S.T. Varum, V.M. Eisinger, D.L. Carlisle, P.J. Donovan, G.P. Schatten, mTOR-mediated activation of p70 S6K induces differentiation of pluripotent human embryonic stem cells, *Cell Reprogram.* 12 (2010) 263–273.
 - [62] J.T. Rodgers, K.Y. King, J.O. Brett, M.J. Cromie, G.W. Charville, K.K. Maguire, C. Brunson, N. Mastey, L. Liu, C.R. Tsai, M.A. Goodell, T.A. Rando, mTORC1 controls the adaptive transition of quiescent stem cells from G0 to G(Alert), *Nature* 510 (7505) (2014) 393–396.
 - [63] C. Chen, Y. Liu, R. Liu, T. Ikenoue, K.L. Guan, Y. Liu, P. Zheng, TSC-mTOR maintains quiescence and function of hematopoietic stem cells by repressing mitochondrial biogenesis and reactive oxygen species, *J. Exp. Med.* 205 (10) (2008) 2397–2408.
 - [64] B. Gan, E. Sahin, S. Jiang, A. Sanchez-Aguilera, K.L. Scott, L. Chin, D.A. Williams, D.J. Kwiatkowski, R.A. DePinho, mTORC1-dependent and -independent regulation of stem cell renewal, differentiation, and mobilization, *Proc. Natl. Acad. Sci. U. S. A.* 105 (2008) 19384–19389.
 - [65] K.A. Overmyer, C.R. Evans, N.R. Qi, C.E. Minogue, J.J. Carson, C.J. Chermiside-Scabbo, L.G. Koch, S.L. Britton, D.J. Pagliarini, J.J. Coon, C.F. Burant, Maximal oxidative capacity during exercise is associated with skeletal muscle fuel selection and dynamic changes in mitochondrial protein acetylation, *Cell Metab.* 21 (3) (2015) 468–478.
 - [66] H. Pilegaard, B. Saltin, P.D. Neuffer, Exercise induces transient transcriptional activation of the PGC-1 α gene in human skeletal muscle, *J. Physiol.* 546 (2003) 851–858.
 - [67] E. Louis, U. Raue, Y. Yang, B. Jemio, S. Trappe, Time course of proteolytic, cytokine, and myostatin gene expression after acute exercise in human skeletal muscle, *J. Appl. Physiol.* 103 (2007) 1744–1751.
 - [68] F. Kadi, P. Schjerling, L.L. Andersen, N. Charifi, J.L. Madsen, L.R. Christensen, J.L. Andersen, The effects of heavy resistance training and detraining on satellite cells in human skeletal muscles, *J. Physiol.* 558 (2004) 1005–1012.
 - [69] A.L. Mackey, B. Esmarck, F. Kadi, S.O. Koskinen, M. Kongsgaard, A. Sylvestersen, J.J. Hansen, G. Larsen, M. Kjaer, Enhanced satellite cell proliferation with resistance training in elderly men and women, *Scand. J. Med. Sci. Sports* 17 (2007) 34–42.
 - [70] M. Kurosaka, H. Naito, Y. Ogura, S. Machida, S. Katamoto, Satellite cell pool enhancement in rat plantaris muscle by endurance training depends on intensity rather than duration, *Acta Physiol (Oxf.)* 205 (2011) 159–166.
 - [71] A.L. Mackey, L. Holm, S. Reitelsheder, T.G. Pedersen, S. Doessing, F. Kadi, M. Kjaer, Myogenic response of human skeletal muscle to 12 weeks of resistance training at light loading intensity, *Scand. J. Med. Sci. Sports* 21 (2011) 773–782.
 - [72] H.K. Smith, T.L. Merry, Voluntary resistance wheel exercise during post-natal growth in rats enhances skeletal muscle satellite cell and myonuclear content at adulthood, *Acta Physiol (Oxf.)* 204 (2012) 393–402.
 - [73] D.A. Hood, Invited review: contractile activity-induced mitochondrial biogenesis in skeletal muscle, *J. Appl. Physiol.* 90 (2001) 1137–1157.
 - [74] M. Flüch, H. Hoppeler, Molecular basis of skeletal muscle plasticity from gene to form and function, *Rev. Physiol. Biochem. Pharmacol.* 146 (2003) 159–216.

- [75] V.G. Coffey, J.A. Hawley, The molecular bases of training adaptation, *Sports Med.* 37 (2007) 737–763.
- [76] J.R. Jackson, J. Mula, T.J. Kirby, C.S. Fry, J.D. Lee, M.F. Ubele, et al., Satellite cell depletion does not inhibit adult skeletal muscle regrowth following unloading-induced atrophy, *Am. J. Phys. Cell Phys.* 303 (8) (2012) C854–C861.
- [77] K. Matsumoto, M. Mizuno, T. Mizuno, B. Dilling-Hansen, A. Lahoz, V. Bertelsen, H. Münster, H. Jordening, K. Hamada, T. Doi, Branched-chain amino acids and arginine supplementation attenuates skeletal muscle proteolysis induced by moderate exercise in young individuals, *Int. J. Sports Med.* 28 (6) (2007) 531–538.
- [78] K. Legerlotz, B. Elliott, B. Guillemin, H.K. Smith, Voluntary resistance running wheel activity pattern and skeletal muscle growth in rats, *Exp. Physiol.* 93 (6) (2008) 754–762.
- [79] J.R. Jackson, T.J. Kirby, C.S. Fry, R.L. Cooper, J.J. McCarthy, C.A. Peterson, E.E. Dupont-Versteegden, Reduced voluntary running performance is associated with impaired coordination as a result of muscle satellite cell depletion in adult mice, *Skelet. Muscle* 16 (2015) 5–41.
- [80] G. Parise, I.W. McKinnell, M.A. Rudnicki, Muscle satellite cell and atypical myogenic progenitor response following exercise, *Muscle Nerve* 37 (5) (2008) 611–619.
- [81] S. Fujimaki, R. Hidaka, M. Asashima, T. Takemasa, T. Kuwabara, Wnt protein-mediated satellite cell conversion in adult and aged mice following voluntary wheel running, *J. Biol. Chem.* 289 (11) (2014) 7399–7412.
- [82] C.S. Fry, J.D. Lee, J. Mula, T.J. Kirby, J.R. Jackson, F. Liu, L. Yang, C.L. Mendias, E.E. Dupont-Versteegden, J.J. McCarthy, C.A. Peterson, Inducible depletion of satellite cells in adult, sedentary mice impairs muscle regenerative capacity without affecting sarcopenia, *Nat. Med.* 21 (1) (2015) 76–80.
- [83] M. Malatesta, P. Fattoretti, M. Giagnacovo, et al., Physical training modulates structural and functional features of cell nuclei in type II myofibers of old mice, *Rejuvenation Res.* 14 (2011) 543–552.
- [84] A. Ishihara, R.R. Roy, Y. Ohira, Y. Ibata, V.R. Edgerton, Hypertrophy of rat plantaris muscle fibers after voluntary running with increasing loads, *J. Appl. Physiol.* 84 (6) (1998) 2183–2189.
- [85] J. Konhilas, U. Widegren, D. Allen, A. Paul, A. Cleary, L. Leinwand, Loaded wheel running and muscle adaption in the mouse, *Am. J. Physiol. Heart Circ. Physiol.* 289 (2005) H455–H465.
- [86] A.S. Brack, I.M. Conboy, M.J. Conboy, J. Shen, T.A. Rando, A temporal switch from Notch to Wnt signaling in muscle stem cells is necessary for normal adult myogenesis, *Cell Stem Cell* 2 (2008) 50–59.
- [87] H.K. Smith, L. Maxwell, C.D. Rodgers, N.H. McKee, M.J. Pyley, Exercise-enhanced satellite cell proliferation and new myonuclear accretion in rat skeletal muscle, *J. Appl. Physiol.* 90 (4) (2001) 1407–1414.
- [88] J.P. Nederveen, S. Joannisse, C.M. Séguin, K.E. Bell, S.K. Baker, S.M. Phillips, G. Parise, The effect of exercise mode on the acute response of satellite cells in old men, *Acta Physiol (Oxf.)* 215 (4) (2015) 177–190.
- [89] F. Macaluso, N.E. Brooks, M. van de Vyver, K. Van Tubbergh, C.U. Niesler, K.H. Myburgh, Satellite cell count, VO(2max), and p38 MAPK in inactive to moderately active young men, *Scand. J. Med. Sci. Sports* 22 (4) (2012) e38–e44.
- [90] S. Joannisse, B.R. McKay, J.P. Nederveen, T.D. Scribbans, B.J. Gurd, J.B. Gillen, M.J. Gibala, M. Tarnopolsky, G. Parise, Satellite cell activity, without expansion, after nonhypertrophic stimuli, *Am. J. Phys. Regul. Integr. Comp. Phys.* 309 (9) (2015) R1101–R1111.
- [91] A.L. Mackey, A. Karlsen, C. Couppé, U.R. Mikkelsen, R.H. Nielsen, S.P. Magnusson, M. Kjaer, Differential satellite cell density of type I and II fibres with lifelong endurance running in old men, *Acta Physiol (Oxf.)* 210 (3) (2014) 612–627.